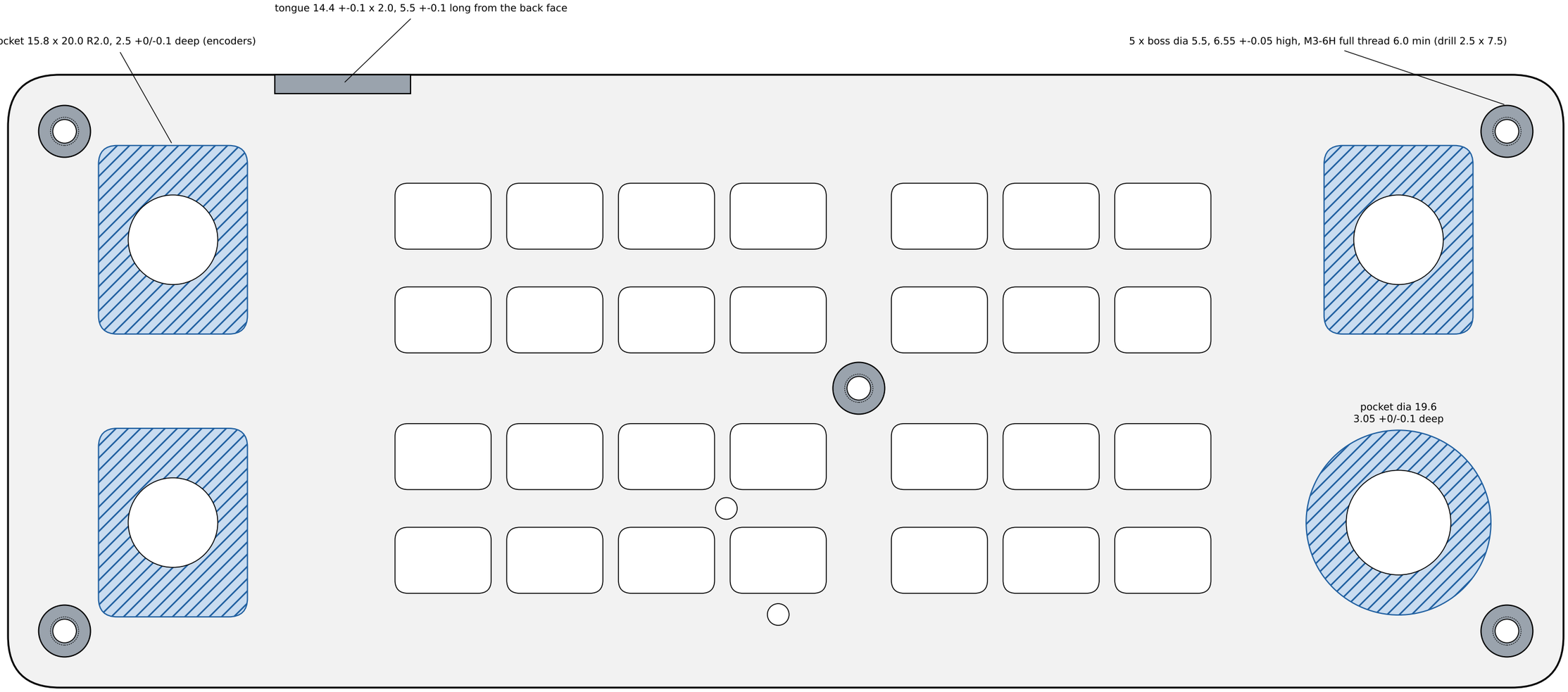
Avionics Panel rev 1 front plate, seen from the BACK (machined side)



Features

key hole	10.2 x 7.0 +0.1/-0 R1.4	through, corner R1.5 max	x28
encoder hole	dia 9.5	through	x3
encoder pocket	15.8 x 20.0 R2.0	2.5 +0/-0.1 deep from back, floor corner R0.3 max	x3
RANGE hole	dia 11.1	through	x1
RANGE pocket	dia 19.6	3.05 +0/-0.1 deep from back, floor corner R0.3 max	x1
SHIFT light hole	dia 2.3 +0.1/-0	through	x1
UNIT light hole	dia 2.3 +0.1/-0	through	x1
boss	dia 5.5, root R0.3 max	6.55 +-0.05 high from back, end face flat	x5
tapped hole	M3-6H in each boss	full thread 6.0 min, drill 2.5 x 7.5 deep	x5
tongue	14.4 +-0.1 x 2.0, root R0.3 max	5.5 +-0.1 long from back, outer face flush with plate edge	x1

Notes

- Material 6061 aluminium, machined from 12 mm plate: the web 4.0 +-0.1 mm thick before anodising, the 5 bosses and the tongue on its back, 10.55 mm high in all (the front may be faced). Outline 165.0 x 65.0 mm, corner R5.5. Volume 32.0 cm3. Units mm, general tolerance ISO 2768-m; flatness 0.1 over the whole web after bead blasting and anodising (the plate is held only at its bosses).
- Front face (+Z in the STEP) is flat: no pockets or fastener holes on the front, and no chamfers beyond the edge break below.
- Positions of the 5 bosses and their tapped holes, of the pockets, of the tongue and of all through holes: +-0.1 to each other (the key caps, the board, the parts in the pockets, the screws and the body's slot must line up).
- Bosses: dia 5.5 +-0.1, 6.55 +-0.05 high from the back face of the web (required: they set the plate height), each end face flat and square to its hole; root R0.3 max.
- Tongue on the back edge: 14.4 +-0.1 wide, 2.0 thick with its outer face flush with the plate's edge, 5.5 +-0.1 long from the back face of the web (it ends short of the bosses); root R0.3 max. It closes the body's slot over the USB-C socket.
- Threads: 5 x M3-6H, one in each boss from its end face, full thread at least 6.0 deep, tap drill dia 2.5 x 7.5 deep plus the drill point, 0.3 x 90 deg countersink: at least 2.15 mm of metal stays in front of the drill point. The threads must gauge M3-6H after anodising: mask them while anodising, or cut them again afterwards. The boss end faces may be anodised or bare, but all of them alike.
- Pocket depths +0, never deeper: at least 0.85 mm of material must stay to the front face under every pocket; nothing may break through or mark the front face.
- Encoder pockets and RANGE pocket: floor corner (inside radius between wall and floor) at most 0.3 mm; the parts' top corners come within about 0.2 mm of the pocket walls.
- Finish: black anodised, matte (bead blasted before anodising); the pockets may be anodised or bare, but all pockets alike (all anodised or all bare).
- Marking on the front per avionics-panel-rev1-plate.dxf, same origin (bottom-left, seen from the front): layer LASER = laser marking, layer UV_AMBER = UV print in JLCCNC's standard orange. Layer CUT = outline and through holes for reference. DXF units mm; fill closed contours even-odd (letter counters are separate contours).
- Key holes 10.2 x 7.0 +0.1/-0 after anodising (never smaller: the key caps centre the board in them), corners R1.4, R1.5 max. Light holes +0.1/-0 (never smaller: the light pipes slide in them). Deburr all edges: edge break at most 0.2 on the front outline and hole edges; on the back edges of the key holes 0.3 to 0.5 (a lead-in for the key caps); the light holes' back edges sharp, deburred only (0.05 max: the light pipes' shoulders bear there).